

Module 5

Troubleshooting Deployment Issues and Cost

Module Objectives

- Troubleshoot cloud deployment and migration issues.
- Understand cloud cost considerations.

Lesson 5.1

Troubleshoot Deployment and Migration Issues

Cloud+ CV0-004 Certification Exam Objective:

- 6.1 Given a scenario, troubleshoot deployment issues.

Troubleshoot Deployment and Migration Issues

- Troubleshoot Deployment Issues
- Compatibility Problems
- Resource Allocation Misconfiguration
- Instance Configuration and Performance Testing
- Permissions and Privilege Misconfiguration
- Troubleshoot Services and Applications
- Deprecation of Functionality
- Cloud Service Outages
- Regional Service Availability

Troubleshoot Deployment Issues

- Identify the problem.
- Determine the scope of the problem.
- Establish a theory of probable cause or question the obvious.
- Test the theory to determine the cause.
- Establish a plan of action.
- Implement the solution or escalate.
- Verify full system functionality.
- Implement preventive measures.
- Perform a root cause analysis.
- Document findings, actions, and outcomes throughout the process.

Compatibility Problems

- Compatibility issues
- Service quotas
- Application programming interface (API) throttling

Resource Allocation Misconfiguration

- Central processing unit (CPU) allocation
- GPU allocation
- Memory allocation
- Storage input/output (I/O) and capacity allocation
- Network allocation

Instance Configuration and Performance Testing

- Instance misconfiguration due to sizing issues
- Failed migration and integration of vendors or platforms due to sizing issues
- Instance misconfiguration
- Failed migration and integration of vendors or platforms
- Service provider oversubscription

Permissions and Privilege Misconfiguration

- Standard users or groups should be authorized to access a resource but cannot
- Administrator cannot log in to a VM instance with admin privileges
- Cannot escalate privileges on the VM instance

Troubleshoot Services and Applications

- Software components are out of date or deprecated
- Services are no longer available
- Infrastructure resources (CPU, memory, storage, networking, etc.) are no longer available

Deprecation of Functionality

- Planned obsolescence in technology
- Hardware and software evolution
- Practical reasons for deprecation
- Notice and migration periods
- Resistance to change
- Financial burden and technological advancement

Cloud Service Outages

- Check for full or partial outage issues
- Mitigate future outage issues

Regional Service Availability

- Infrastructure availability
- Geopolitical concerns
- Services or options not available in the given region

Lesson 5.2

Cost Considerations Related to Cloud Usage

Cloud+ CV0-004 Certification Exam Objective:

- 1.8 Summarize cost considerations related to cloud usage.

Cost Considerations Related to Cloud Usage

- Subscription Services Costs
- Price Calculators and Cost Analysis
- Cloud Billing Models
- Resource Allocation
- Resource Metering
- Resource Tagging
- Rightsizing
- Troubleshoot Billing

Subscription Services Costs

- Licensing model with regular fees (monthly/annually) for access to resources
- Web-based interfaces for services, costs, and agreements (e.g., GCP Cloud Billing)
- Shared responsibility between cloud admins and financial departments
- Cost comparison

Price Calculators and Cost Analysis

- Price estimators
- Managed CSPs
- Discounts

Cloud Billing Models

- Dedicated host with VM instance
- On-demand instance (pay-as-you-go)
- Reserved instance
- Spot instance

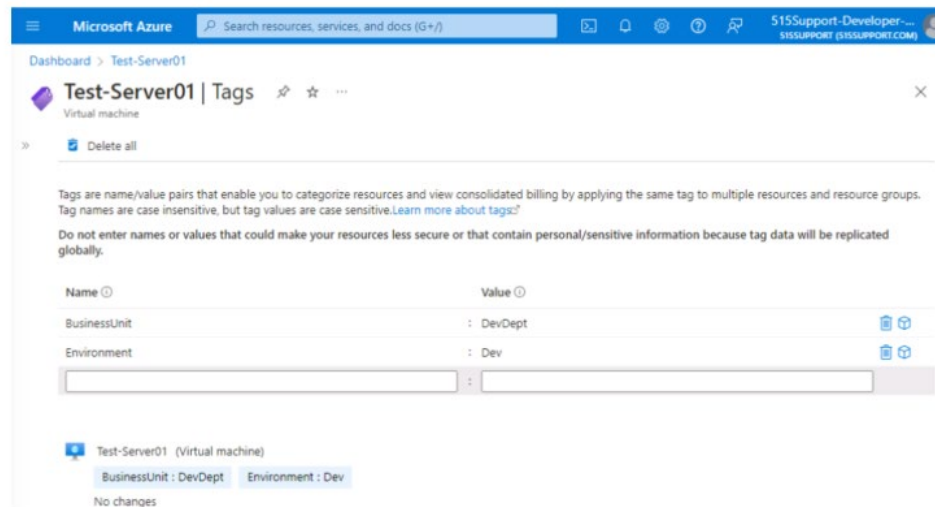
Resource Allocation

- Resource metering
- Tagging
- Rightsizing

Resource Metering

- Benefits
 - Cost optimization
 - Tagging
 - Utilization reports
- Key Points
 - Billing models
 - Rightsizing
 - Auto-scaling

Resource Tagging



This VM has two resource tags - DevDept and Dev. (Screenshot courtesy of Amazon.)

Rightsizing

Viewing 207 of 695 available instances

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Instance name ▲	On-Demand hourly rate ▼	vCPU ▼	Memory ▼	Storage ▼	Network performance ▼
t2.nano	\$0.0058	1	0.5 GiB	EBS Only	Low
t2.micro	\$0.0116	1	1 GiB	EBS Only	Low to Moderate
t2.small	\$0.023	1	2 GiB	EBS Only	Low to Moderate
t2.medium	\$0.0464	2	4 GiB	EBS Only	Low to Moderate
t2.large	\$0.0928	2	8 GiB	EBS Only	Low to Moderate
t2.xlarge	\$0.1856	4	16 GiB	EBS Only	Moderate
t2.2xlarge	\$0.3712	8	32 GiB	EBS Only	Moderate

AWS general purpose on-demand instances. Note the price versus performance information. (Screenshot courtesy of Amazon.)

Troubleshoot Billing

- Technical support is unavailable or unsuitable
- Service availability is less than required
- Licensing issues
- Cost/billing issues

Module Summary

- Identify and troubleshoot provisioning and migration issues.
- Cloud services shift IT costs to subscription fees.
- Understand billing models and resource configuration.
- Optimize resource utilization to manage expenses.